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Advanced Track Inspection Technology Demonstration Platform

Driving Innovation Through Collaboration: Enhancing Safety and Operational Efficiency

DALLAS, TX



The NRC at a Glance



We are the Government of Canada's
largest research and development organization
with over 100 years of experience

Mandate...

- Advance scientific and technical knowledge
- Support government policy objectives
- Support business innovation



Presentation Outline

- Motivation
- Vision & Approach
- Project
 - Overview – Hardware
 - Overview – Data Integration and Analytics
 - Overview – Data Confidentiality Management
 - Timeline
- Case Studies
 1. Impact of friction management on lateral force
 2. Wheel climbing risk
 3. Combined track geometry effect
 4. Carbody motion indicating track surface conditions
 5. Rail surface defects identified by wheel impact acceleration
 6. Brake application and in-train force
- Next Steps
- Conclusion





The way forward:



Collaboration among...



To:

- Remove adoption barriers.
- Improve efficiency without compromising safety.

Motivation

Adopting advanced technologies **improves** rail safety and efficiency, but **requires** rigorous testing before deployment.

Challenges:

- **Shortlines** lack resources to assess and apply new technologies.
- **Technology developers/suppliers** have limited opportunities for real-world demonstrations.
- **Single-supplier** demo projects are cost-prohibitive.
- **Academia** lacks access to operational data for research and training.
- **Railways** face too many new technologies — selection and integration are difficult.
- Concerns about sharing confidential or proprietary data.
- Integrating and interpreting large volumes of data from multiple systems is complex.



Vision & Approach

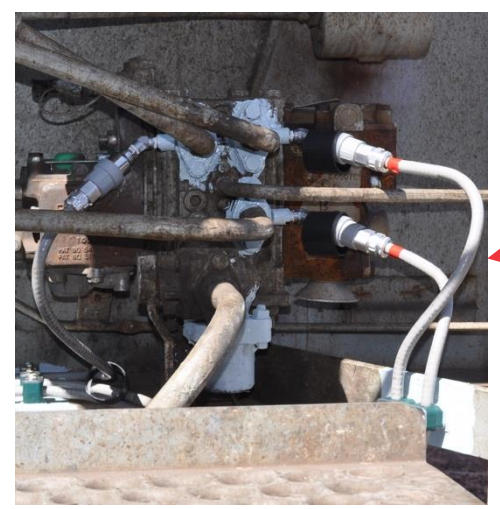


- NRC and Transport Canada collaborating to collect track performance data using a revenue-service freight railcar on shortline networks.
- Project uses new technology for continuous track monitoring to improve efficiency, safety, and support regulatory and policy advancement.

Project Overview - Hardware



Loaded the railcar to a GRL of 259,500 lb with lightweight vesicular basalt.



Brake System Monitoring



Power Management System



Accelerometers



Antennas



Weather Station



Data Acquisition and Transmission



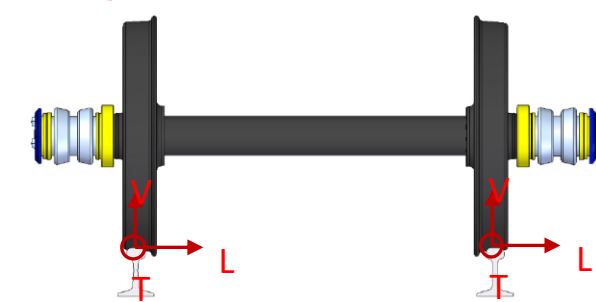
Instrumented Coupler



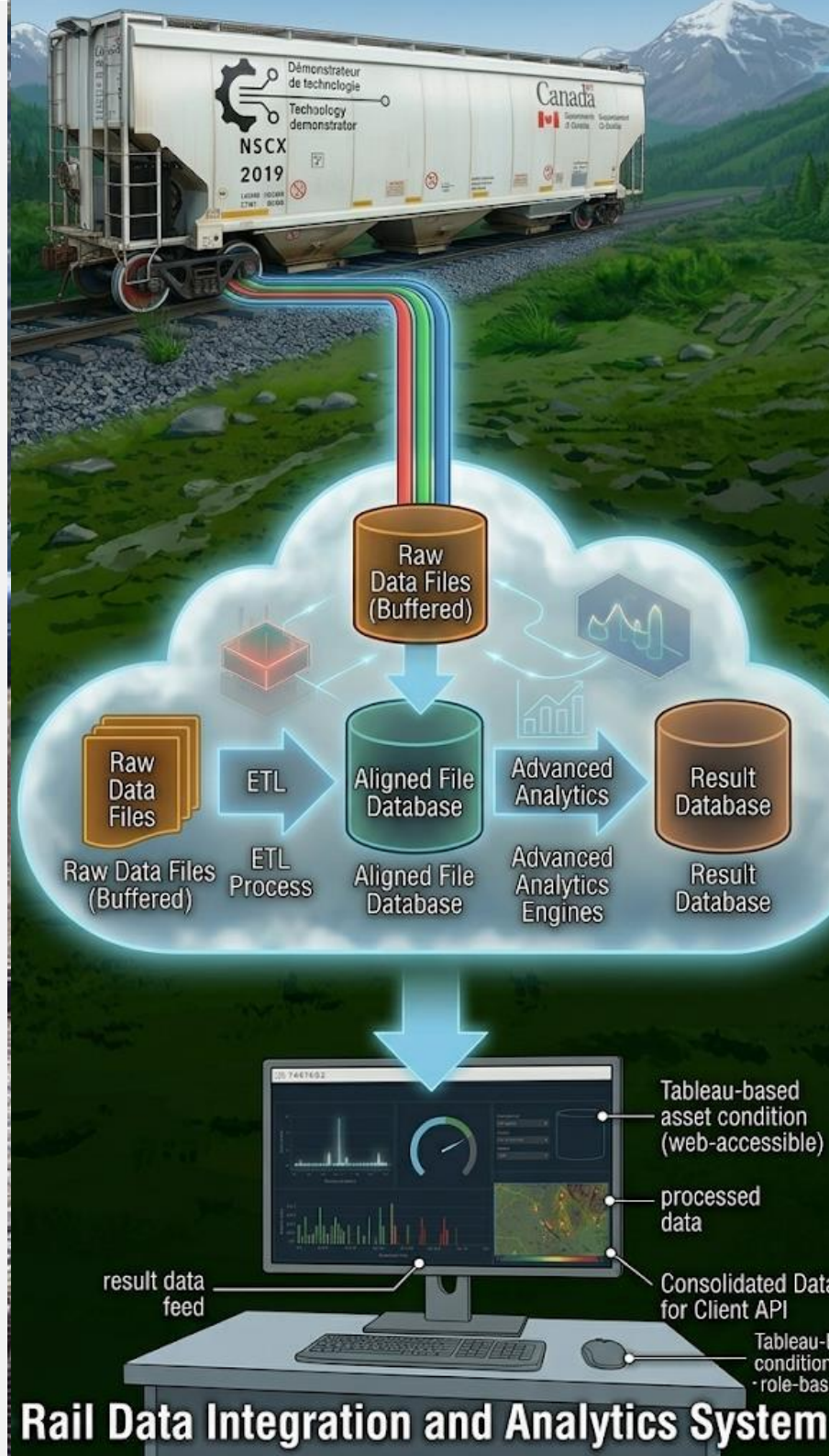
Modified the railcar structure to accept the ATGMS.



ATGMS



Two Instrumented Wheelsets



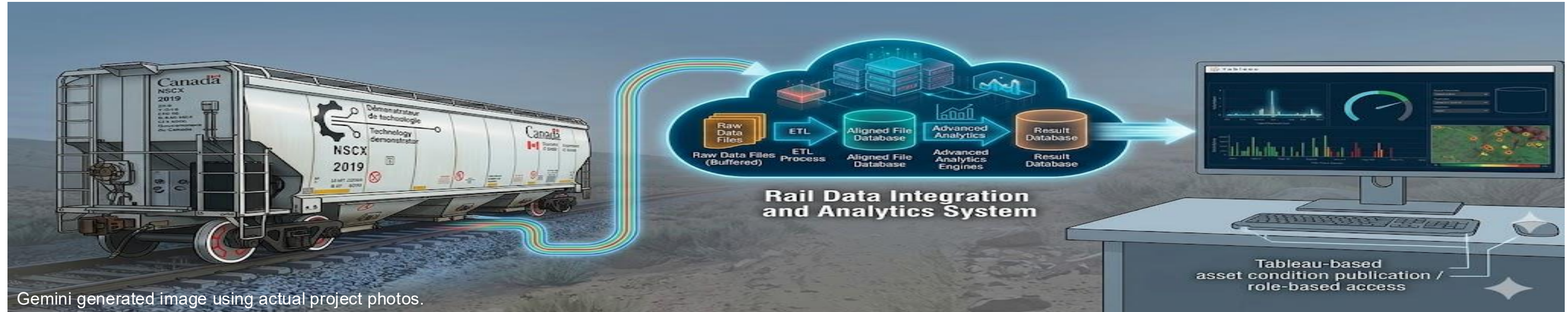
Gemini generated image using actual project photos

Project Overview - Data Integration and Analytics

- **Rail Data Integration and Analytics System (RDIAS)**; an automatic data management, processing and analytics system.
- **RDIAS includes the following components:**
 - ETL (extraction, transformation and loading);
 - Multi-dimensional databases;
 - Advanced data analytics algorithms;
 - Result visualization on dashboards.



Project Overview - Data Integration and Analytics Cont.



Platform Sensor Data:

- Track geometry;
- IWS forces (vertical, lateral, long);
- Car body motion;
- Wheel impacts;
- In-train (coupler) force;
- Brake system pressure;
- Temperature, weather.

Railway Operator Data:

- Historical defects;
- Track features & infrastructure;
- Other.

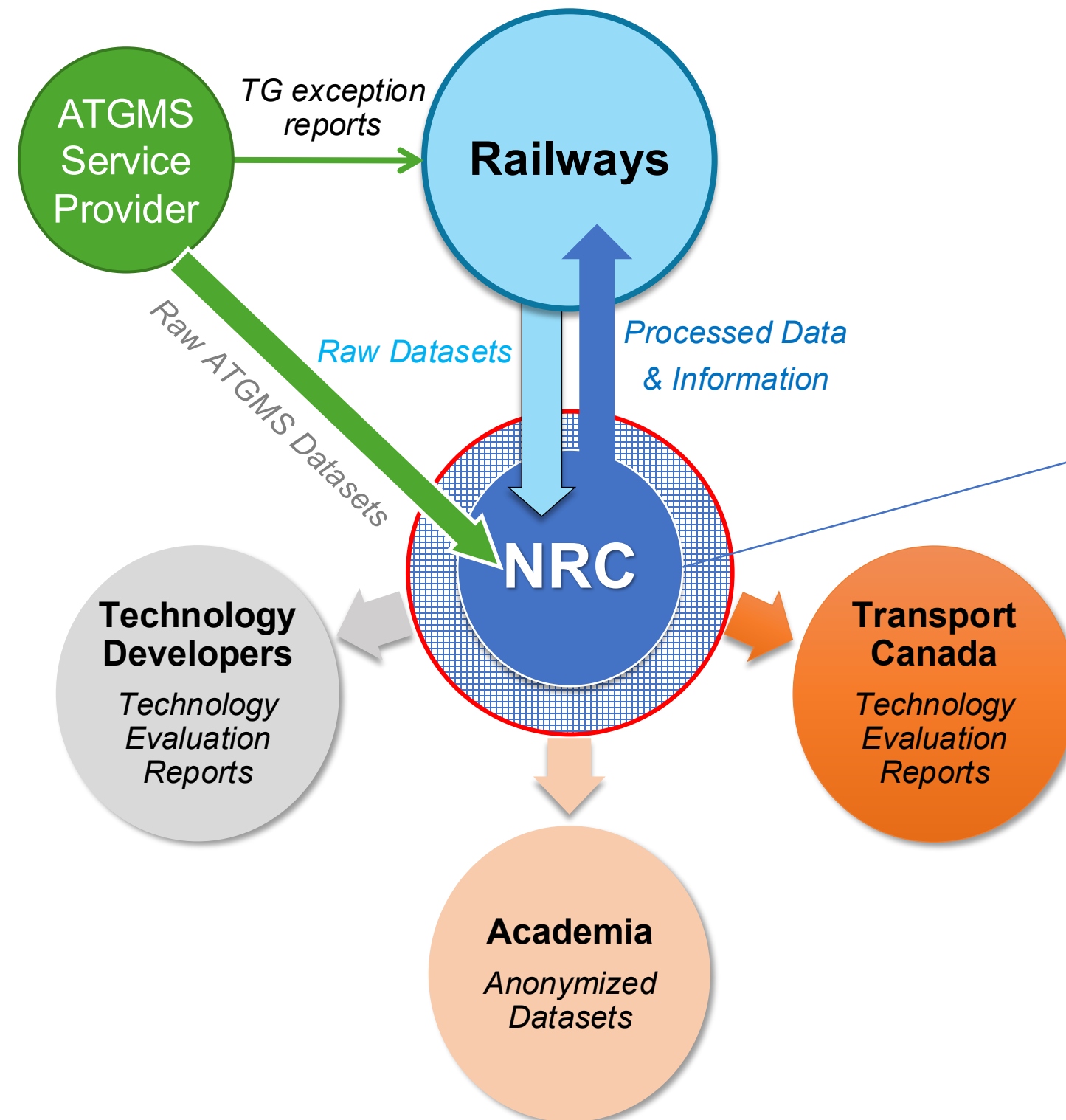
Foot by Foot Drill Down Screen



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Project Overview - Data Confidentiality Management



- Independent / impartial technology evaluation partner;
- Analyze datasets;
- (Anonymized) data sharing and confidentiality agreements in place with stakeholders;
- Obtain approval from railways prior to any results dissemination (presentations, reporting).

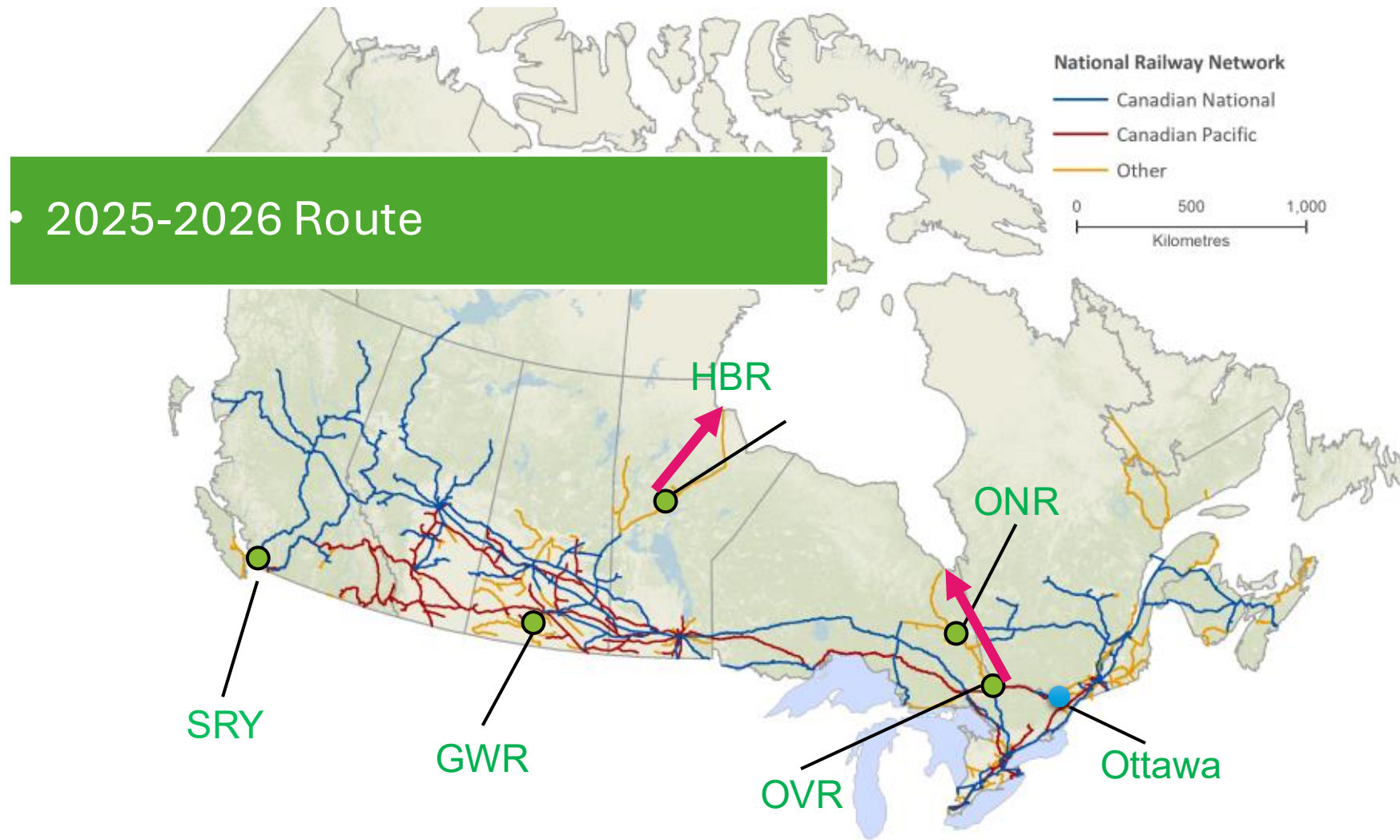
- **Anonymized data only.** All railway identifiers (GPS locations, track charts, rail profiles, etc) will be removed from any data reported and/or disseminated to stakeholders.
- No railway TG exception reports are shared with regulator.
- NRC does not receive/have access to TG exception reports.



Timeline



Deployment and Operations



Shortline Railway	Data Collected (Miles)
ONR: Ontario Northland Railway	1,690
OVR: Ottawa Valley Railway	500
HBR: Hudson Bay Railway	1,300
GWR: Great Western Railway	300
SRY: Southern Railway of British Columbia	128
Total:	3,918

Supports National Projects: Northlander and Port of Churchill

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Testimony of Impact

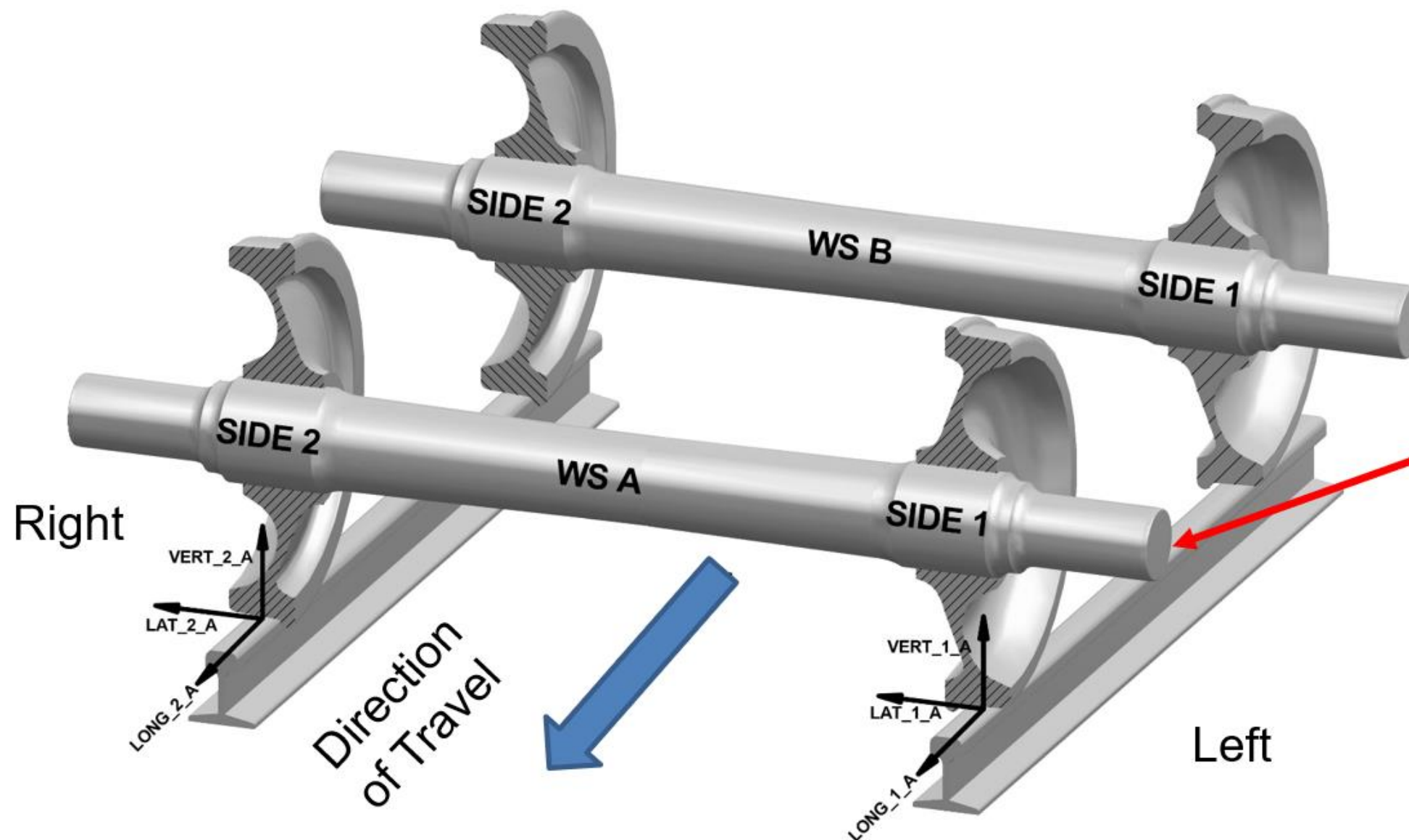
- "The TDP has helped us identify issues that we might not have known about until they became disruptive. The ability to identify issues early will allow us to resolve them before they become safety concerns."
- "... use of the TDP will allow us to monitor our track by quantifying defect growth rates, identifying trends and manage risk."

Brett Young, General Manager,
Hudson Bay Railway

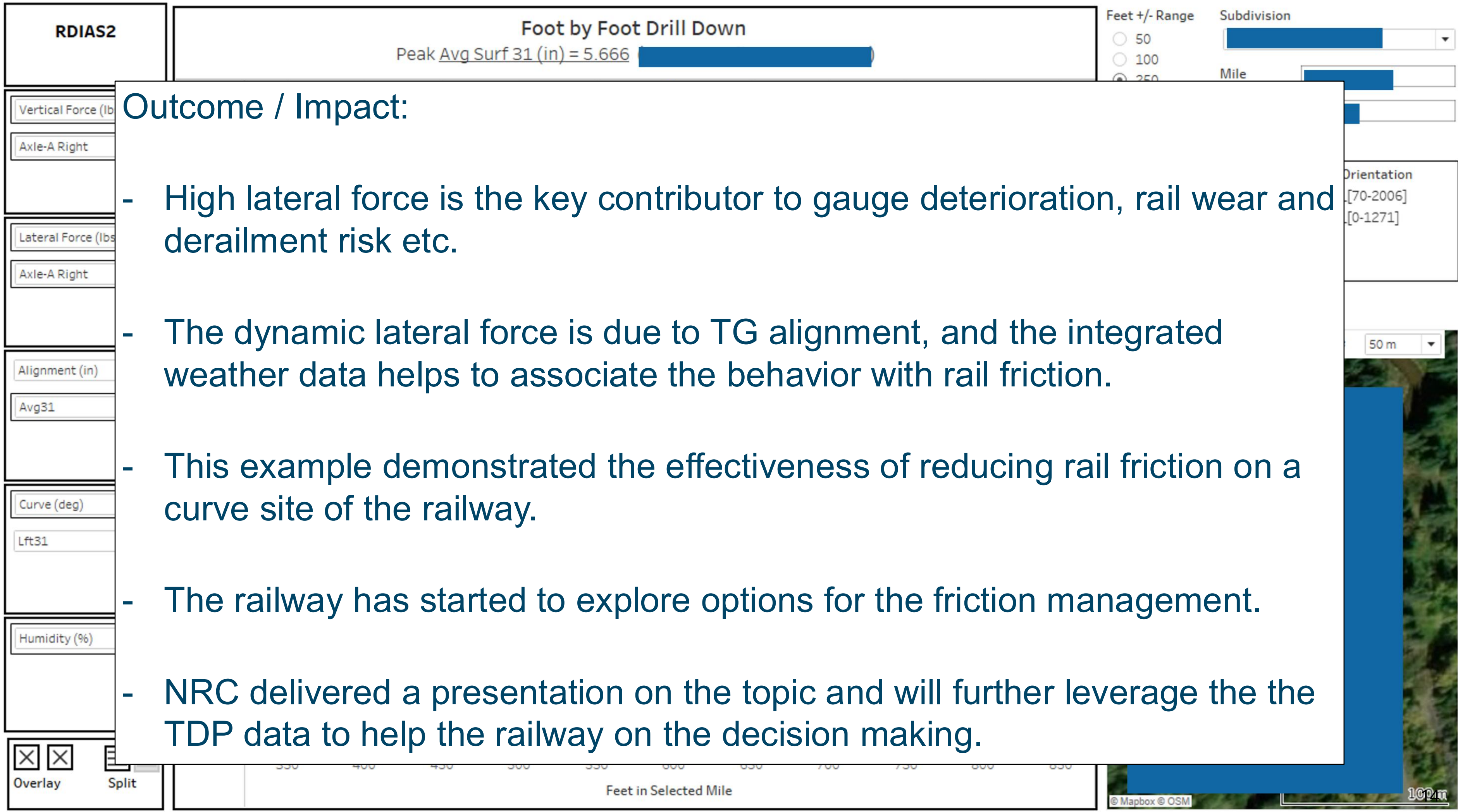


Instrumented Wheelset

- Measure wheel – rail force in three directions
- Identify “pain” points and potential risks

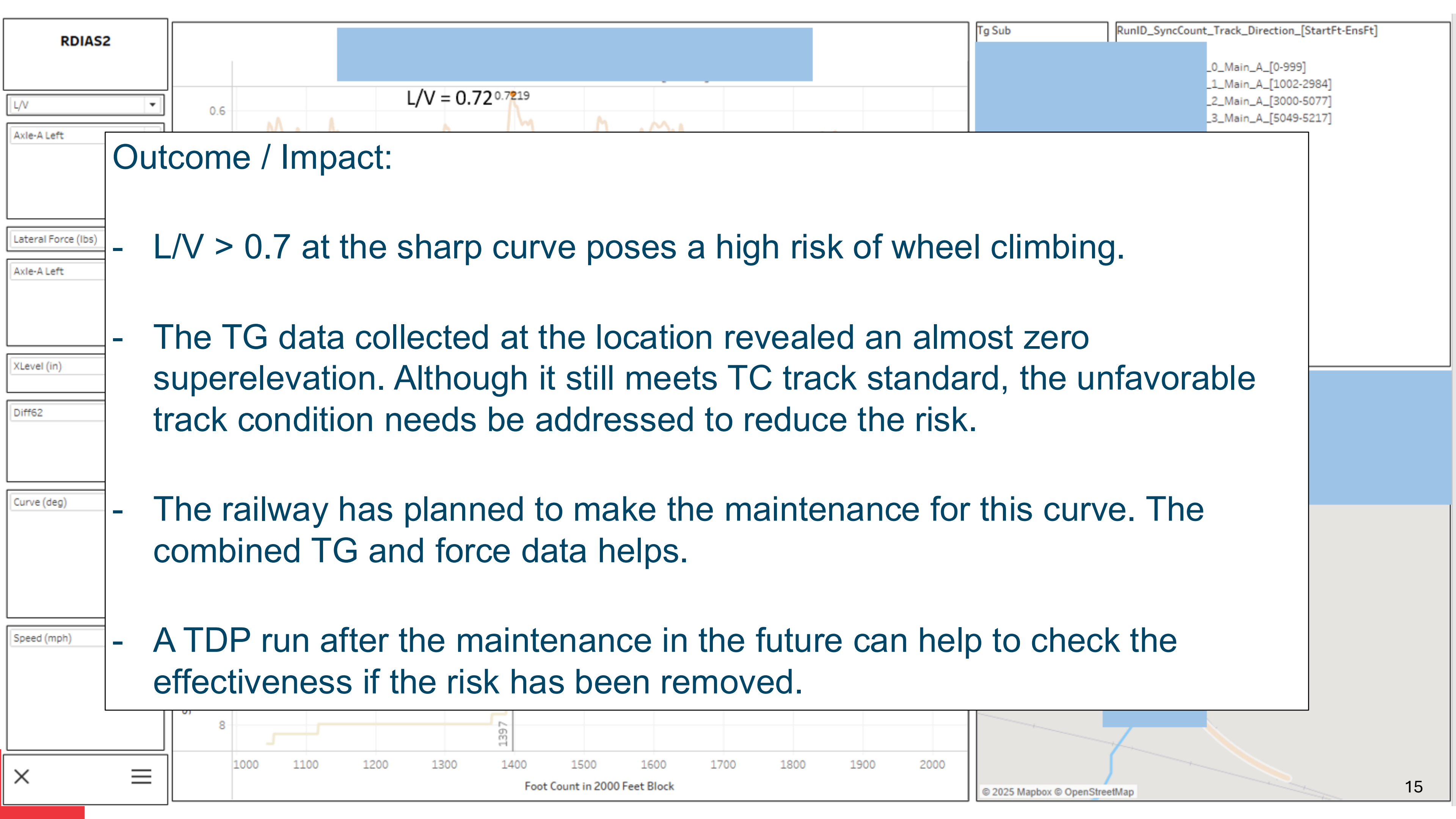


- Truck Side Vertical Force (Plate cut)
- Truck Side Lateral Force (Wide gauge)
- Wheel L/V Ratio (Wheel climbing risk)
- Truck Side L/V Ratio (Rail rollover risk)



Outcome / Impact:

- High lateral force is the key contributor to gauge deterioration, rail wear and derailment risk etc.
- The dynamic lateral force is due to TG alignment, and the integrated weather data helps to associate the behavior with rail friction.
- This example demonstrated the effectiveness of reducing rail friction on a curve site of the railway.
- The railway has started to explore options for the friction management.
- NRC delivered a presentation on the topic and will further leverage the the TDP data to help the railway on the decision making.

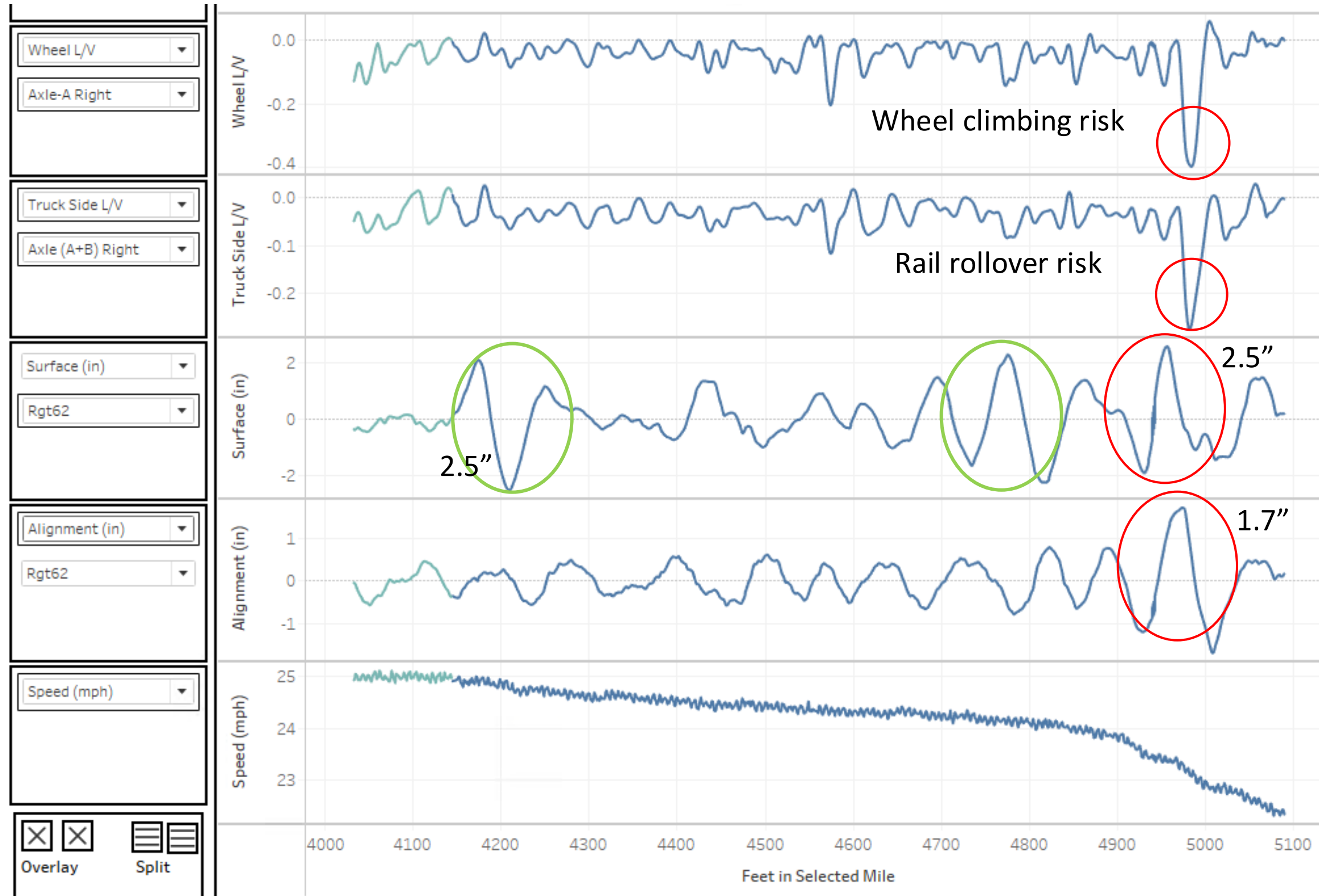


Outcome / Impact:

- $L/V > 0.7$ at the sharp curve poses a high risk of wheel climbing.
- The TG data collected at the location revealed an almost zero superelevation. Although it still meets TC track standard, the unfavorable track condition needs be addressed to reduce the risk.
- The railway has planned to make the maintenance for this curve. The combined TG and force data helps.
- A TDP run after the maintenance in the future can help to check the effectiveness if the risk has been removed.



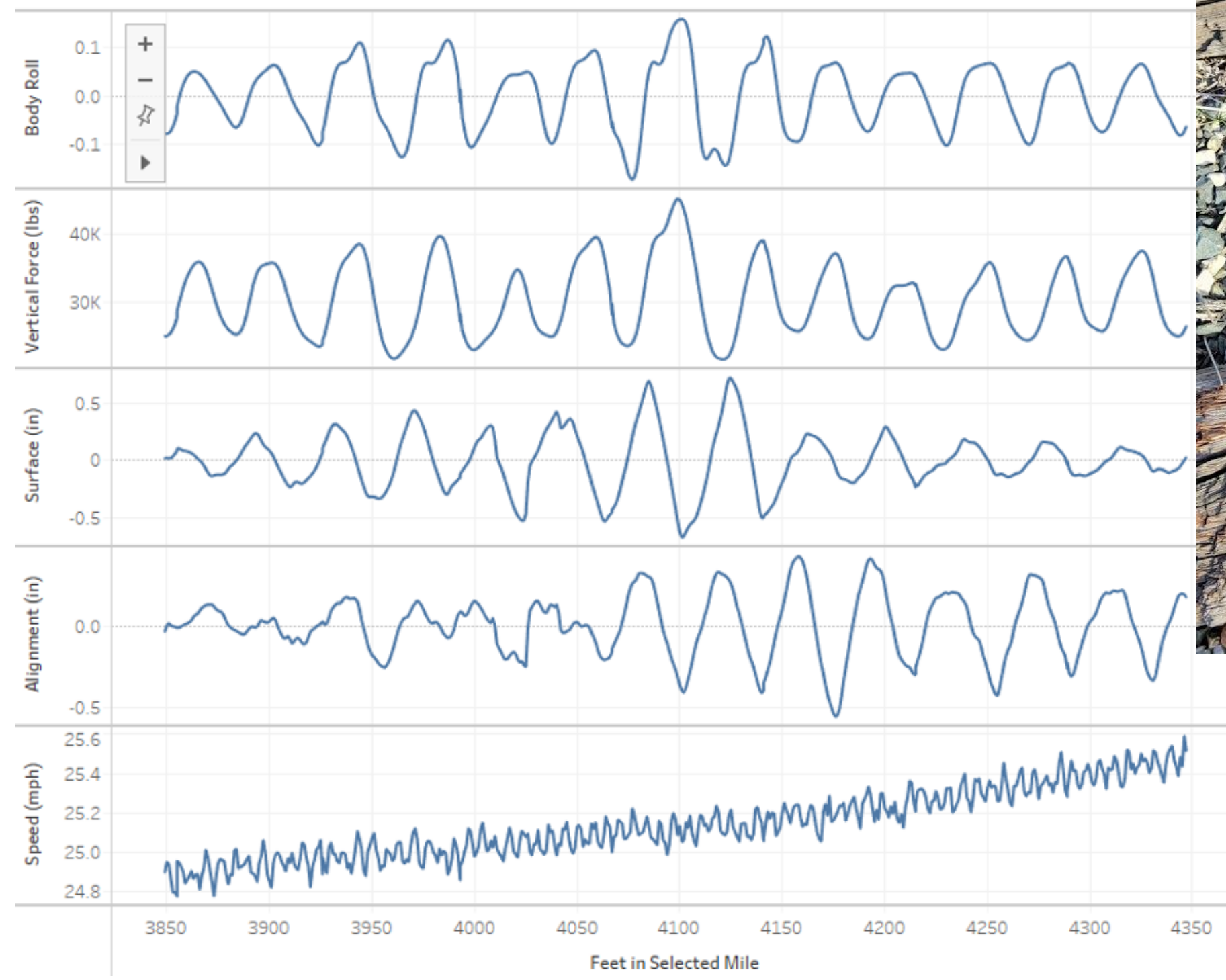
Combined Track Geometry associated High Risk Track Location



- TDP conducts synched force / TG measurement that can provide valuable dataset for determining the high risk combinations of track geometries.
- This example shows that two single surface 62 peaks (without accompanying alignment peak) resulted much lower derailment risk than the combined case.
- The RDIAS dashboard is an effective tool to review force and TG data together. It can help railway to make maintenance planning decision such as:
 - which of the three near urgent surface defects should be addressed first?



Carbody Motion, Vertical Force, Track Surface and Field Conditions



Surface conditions unknown to the railway



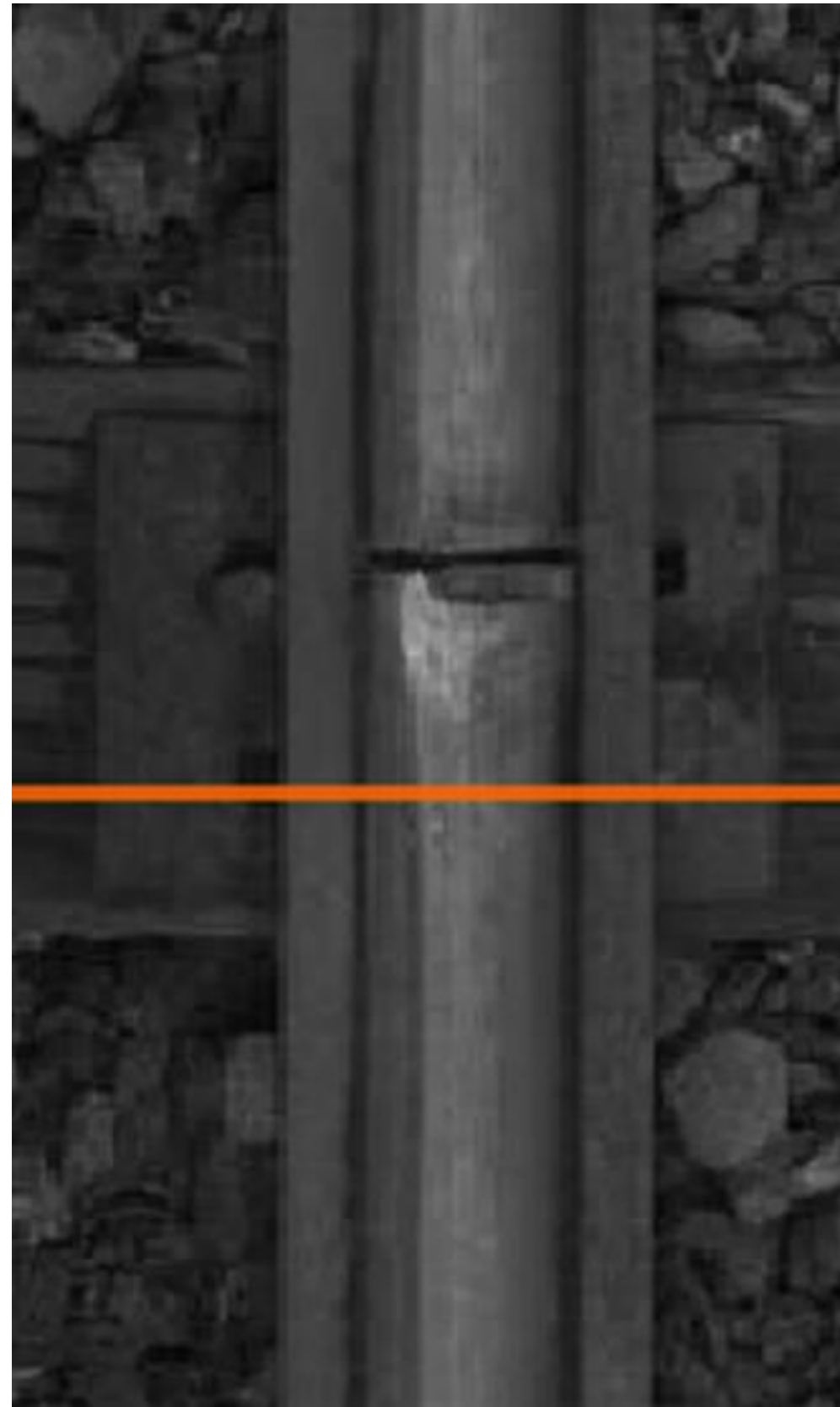


High Wheel Impact Acceleration Events Aligned with Track Image Measurement

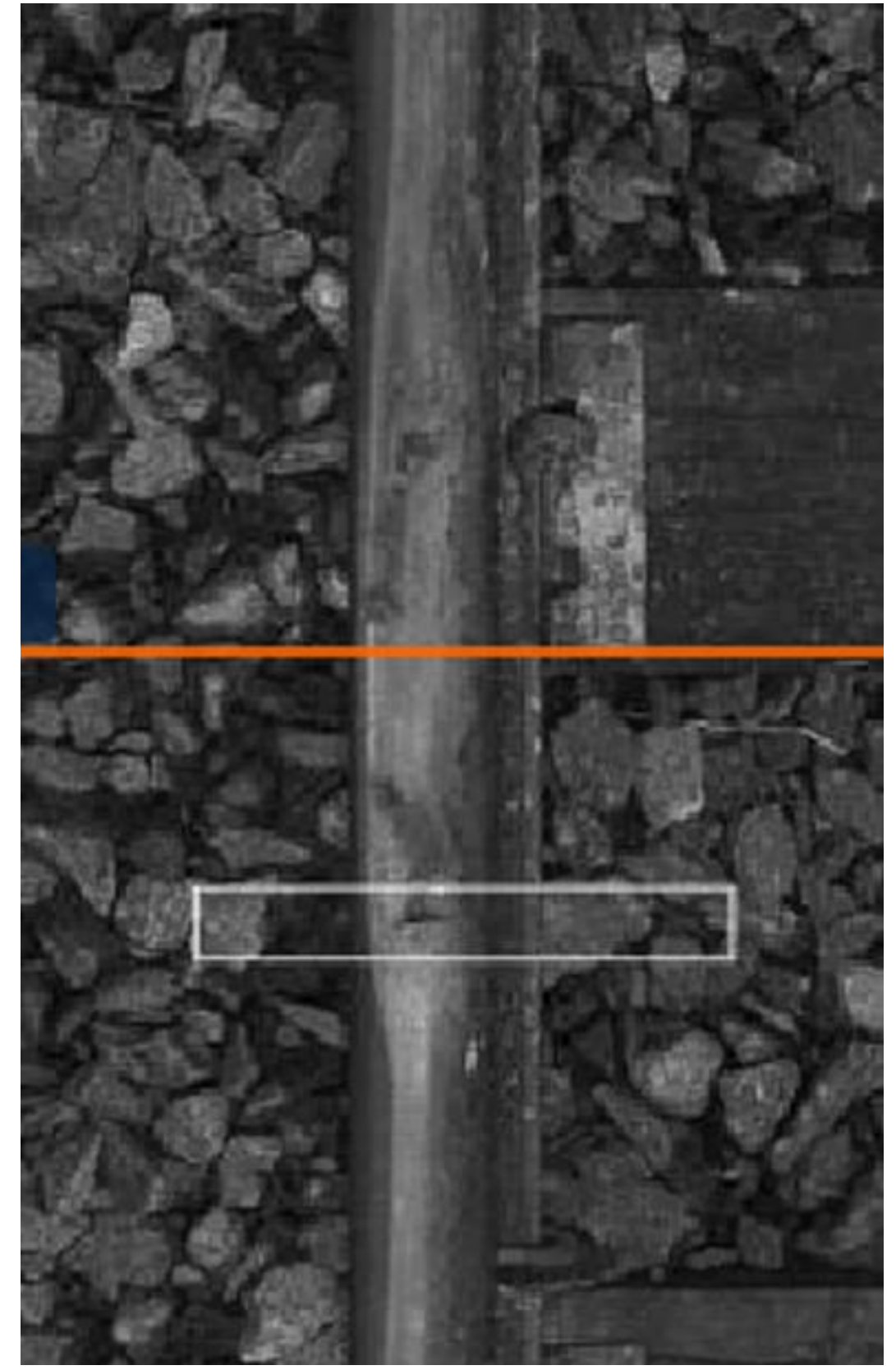
59g Impact (joint)



64g Impact (joint)

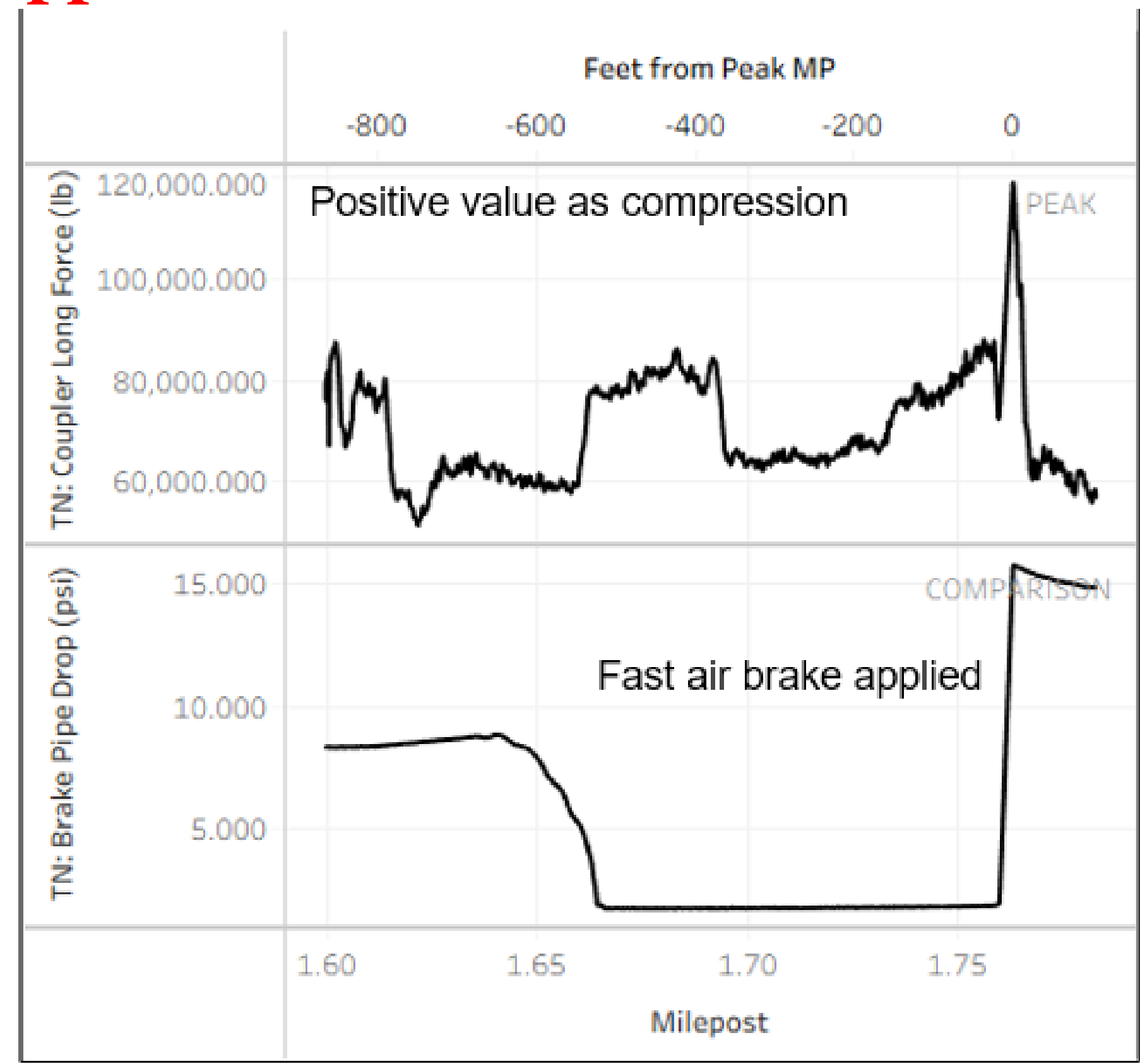


53g Impact (non-joint)





An Example of High In-Train Force Caused by Rapid Air Brake Application



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Some Other Findings and Examples

- Both alignment and lateral forces were found to be high at a bridge location. A manual inspection conducted during a train passage **revealed some structural deflection beneath the bridge**, an issue previously unknown to the railway. The bridge is now being monitored by the railway for further assessment.
- RDIAS flagged a **high lateral force** on the spiral section of a mild curve. Drilldown analysis identified co-existing **Warp 31 and reverse elevation defects** within the track section. Based on the information, the railway immediately implemented a slow order to mitigate the operation risk.
- NRC and a shortline railway conducted an inspection for a turnout identified by high wheel climbing risk. **A condition known as wheel flange back contact was detected**, and the wheel running marks indicated a potential **risk of low-speed derailment**. The team recommended that the shortline railway apply guard rail lubrication to reduce this risk.
- Several **sharp curves** were flagged based on **high L/V ratio** measurements from the TDP system. Subsequent drilldown analysis by RDIAS confirmed that these sites exhibit combined track geometry conditions, indicated by both ATGMS data and/or car body motion measurements. These locations have been suggested to the railway for further inspection.



Next Steps

Platform Improvement:

- Integrate new sensors (machine vision, rail profiling, GPR, ultrasonic).

Deployment & Data Collection:

- Expand deployment & collect diverse data across varied track conditions.

Data Processing & Validation:

- Use machine learning procedures to automatically remove bad or noisy data.
- Validate track failure diagnosis algorithms to issue alerts for accuracy and operational suitability.
- Enhance RDIAS platform capabilities by incorporating AI capabilities.
- Establish performance metrics for system integration and degradation modeling.

Collaboration & Reporting:

- Document findings and lessons learned in support of project objectives and regulatory modernization goals.





Conclusion

- Successful collaboration among industry, academia, regulators and research organization.
- Effective management of data confidentiality.
- Successful validation of integrated technologies through identification of previously unknown track conditions.
- Demonstrated the value of data-driven decision-making through practical examples.



Collaborators



NATIONAL STEEL CANADA



a Genesee & Wyoming Company

